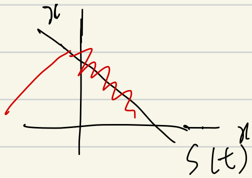
Ch 7. Sliding Control

§7.2 Cont. Approx. of Switching Control laws.

ODE + control "u"

We want to make $\mathbf{x}(t) = \mathbf{x}_d(t)$

Review + what we will going to do.



$s = \dot{\tilde{x}} + \lambda \tilde{x}$

* How to remove chattering

* Trade off between ε & Φ

s.t $|x - x_d| \leq \varepsilon$

Def.

$\tilde{\mathbf{x}}(t) = \mathbf{x}(t) - \mathbf{x}_d(t) = [\tilde{x}, \dots, \tilde{x}^{(n-1)}]$: Tracking error

$s(\mathbf{x}, t) = \left(\frac{d}{dt} + \lambda\right)^{n-1} \tilde{x}$

$S(t) := \{ \mathbf{x} \in \mathbb{R}^n \mid s(\mathbf{x}, t) = 0 \}$

$\frac{1}{2}\frac{d}{dt} s^2 \leq -\eta |s| \iff \dot{s} \leq -\eta \,\mathrm{sgn}(s)$

: Sliding condition

$\Downarrow$

$t_{reach} \leq \frac{|s(t=0)|}{\eta}$

$$\begin{cases} \ddot{x} = f + u \\ |f - \hat{f}| \le F \end{cases}$$

$n = 2$ case

$s(x, t) = \dot{\tilde{x}} + \lambda \tilde{x}$

$$\dot{s} = \ddot{\tilde{x}} + \lambda \tilde{x}$$

$$= f + u - \ddot{x}_d + \lambda \tilde{x}$$

$$u_{eq} = -f + \ddot{x}_d - \lambda \tilde{x}$$ (we don't know f)

$$\hat{u} = -\hat{f} + \ddot{x}_d - \lambda \tilde{x}$$

$$u := \hat{u} - k \operatorname{sgn}(s)$$
$$= -\hat{f} + \ddot{x}_d - \lambda \tilde{x} - k \operatorname{sgn}(s)$$

$$\Rightarrow \frac{1}{2} \frac{d}{dt} s^2 = s \dot{s} = s\left(f - \hat{f} - k \operatorname{sgn}(s)\right)$$
$$\le -\eta |s|$$ (we want to make)

Choose $k := F + \eta$

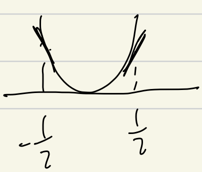
Model imprecision, disturbance $\Rightarrow$ Chattering

$\operatorname{sgn}(s)$: dis continuous

Motivation

$\min f(x)$

$f(x) = x^2$



$x_{n+1} = x_n - k \nabla_x f(x_n)$

$k = 1$

$x_1 = \frac{1}{2}$ $\quad x_2 = -\frac{1}{2}$ $\quad x_3 = \frac{1}{2}$ ...

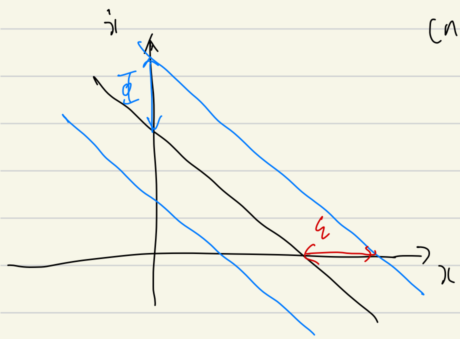
$\ddot{x} = f + bu.$

	Φ	uncertainty
①	constant	f
②	$\Phi(t)$	f
③	$\Phi(t)$	f, b.

1. How to eliminate chattering.
2. Trade off between precision & bandwidth.

Def. Boundary layer

$$B(t) := \{ x \in \mathbb{R}^n \mid |s(x,t)| \le \Phi \}, \quad \Phi > 0$$



$(n=2)$

$$\dot{\tilde{x}} + \lambda \tilde{x} = s(x,t)$$

$$\varepsilon = \frac{\Phi}{\lambda}$$

$$\varepsilon = \frac{\Phi}{\lambda^{n-1}}$$

(Review)

$$|s(t)| \le \Phi. \Rightarrow \tilde{x}^{(i)}(t) \le (2\lambda)^i \frac{\Phi}{\lambda^{n-1}} = (2\lambda)^i \varepsilon.$$

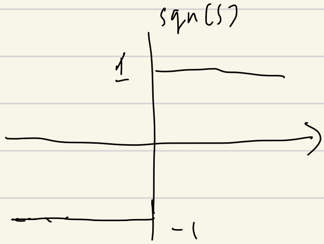
$(\because)$ $\left(\frac{d}{dt}+\lambda\right)\tilde{x}_1 = s \quad \Rightarrow \quad |\tilde{x}_1| \le \frac{\Phi}{\lambda}$

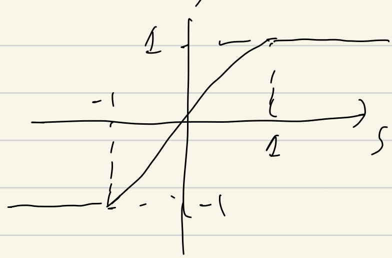
① Remove Chattering

(Same example)

$$\dot{s} = f + u - \ddot{x}_d + \lambda \dot{\tilde{x}}$$

$$u_1 := \hat{u} - k\,\mathrm{sat}\left(\frac{s}{\Phi}\right)$$





(inside band ⇒ decrease the size)

$$\Rightarrow \frac{1}{2}\frac{d}{ds} s^2 = s\left(f - \hat{f} - k\,\mathrm{sat}\left(\frac{s}{\Phi}\right)\right)$$

$$\le \begin{cases} -\eta|s| & s \ge \Phi \\ s\left(F - (F+\eta)\frac{s}{\Phi}\right) & s < \Phi \end{cases}$$

$(k := F+\eta)$

(참고 Fig 7.7, Fig 7.8 Chattering removed, error increased)

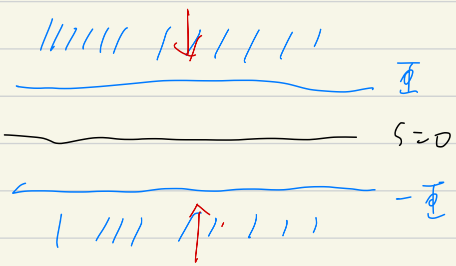
- We need to consider another sliding condition to attract inside of band.
- We want to see the relation between Φ & ε

② $\Phi(t)$: time dependent.

$$s \ge \Phi \Rightarrow \frac{d}{dt}(s-\Phi) \le -\eta$$

$$s \le -\Phi \Rightarrow \frac{d}{dt}(s-(-\Phi)) \ge \eta$$

same



$$|s| \ge \Phi \Rightarrow \frac{1}{2}\frac{d}{dt}s^2 \le (\dot{\Phi}-\eta)|s|$$

$$\dot{s} = f + u - \ddot{x}_d + \lambda\dot{\tilde{x}}$$

$$u_2 := \hat{u} - \bar{K}\,\mathrm{sat}\left(\frac{s}{\Phi}\right) \qquad \bar{K} := K - \dot{\Phi}$$

$$= \hat{u} - (K - \dot{\Phi})\,\mathrm{sat}\left(\frac{s}{\Phi}\right)$$

$$\Rightarrow \quad \frac{1}{2}\frac{d}{dt}s^2 = s\left(f - \hat{f} - \left(k - \dot{\Phi}\right)\mathrm{sat}\left(\frac{s}{\Phi}\right)\right)$$

$\quad k := F + \eta$

$$\le \quad (\dot{\Phi} - \eta)\,|s| \qquad (|s| \geq \Phi)$$

when $\quad |s| \le \Phi$

$$\dot{s} = \underbrace{f - \hat{f}}_{:= -\Delta f} - \bar{k}(x)\frac{s}{\Phi}$$

Assume Δf, $\bar{k}(x)$ smooth (differentiable.)

$$= -\bar{k}(x_d)\frac{s}{\Phi} + \left(-\Delta f(x_d) + O(\varepsilon)\right)$$

$$(\because) \quad \bar{k}(x) - \bar{k}(x_d) = \bar{k}'(x_c)(x - x_d) \le C\varepsilon$$

$$\Delta f(x) - \Delta f(x_d) = \Delta f'(x_c)(x - x_d) \le C\varepsilon$$

(s 에 대한 decaying estimate 를 얻었다.)

(s decay $+$ $|s| \le \Phi \Rightarrow |x - x_d| \le \dfrac{\Phi}{\lambda^{n-1}}$)

Let $\quad \dfrac{\bar{k}(x_d)}{\Phi} := \lambda = \dfrac{k(x_d) - \dot{\Phi}}{\Phi}$ $\quad$ (Balance condition)

$$\dot{\Phi} + \lambda\Phi = k(x_d)$$

(desired time history of boundary layer thickness Φ)

③ $$\ddot{x} = f + bu$$

$$0 < b_{m} \le b \le b_{Mx} \qquad \hat{b} := (b_{m} b_{Mx})^{\frac{1}{2}}$$

$$\Rightarrow \quad \frac{1}{\beta} = \sqrt{\frac{b_{m}}{b_{Mx}}} \le \frac{\hat{b}}{b} \le \sqrt{\frac{b_{Mx}}{b_{m}}} := \beta$$

좋은것

$$\frac{1}{\beta} \le \frac{b}{\hat{b}} \le \beta$$

(Review)

$$\dot{s} = \ddot{\tilde{x}} + \lambda \dot{\tilde{x}} = f + bu - \ddot{x}_d + \lambda \dot{\tilde{x}}$$

$$u_{eq} = b^{-1}(-f + \ddot{x}_d - \lambda \dot{\tilde{x}})$$

$$u := \hat{b}^{-1}(\hat{u} - k\,\mathrm{sgn}(s))$$

$$= \hat{b}^{-1}(-\hat{f} + \ddot{x}_d - \lambda \dot{\tilde{x}} - k\,\mathrm{sgn}(s))$$

$$\Rightarrow \quad \dot{s} = \left(f - \frac{b}{\hat{b}}\hat{f}\right) + \left(1 - \frac{b}{\hat{b}}\right)(-\ddot{x}_d + \lambda \dot{\tilde{x}}) - \frac{b}{\hat{b}} k\,\mathrm{sgn}(s)$$

Now $\Phi \to \Phi(t)$, $\mathrm{sgn}(s) \to \mathrm{sat}\left(\frac{s}{\Phi}\right)$

we want to make.

$$\frac{1}{2}\frac{d}{dt}s^2 \le (\dot{\Phi} - \eta)|s| \qquad \Phi \le |s| \ + \ \mathrm{sat}.$$

$$\dot{s} \le (\dot{\Phi} - \eta)\,\mathrm{sat}\left(\frac{s}{\Phi}\right)$$

$$u_3 := \hat{b}^{-1}\left(-\hat{f} + \ddot{x}_d - \lambda\dot{\tilde{x}} - \bar{k}\,\mathrm{sat}\left(\frac{s}{\Phi}\right)\right)$$

$$\bar{k}(x) = \begin{cases} k(x) - \dfrac{\dot{\Phi}}{\beta} & \text{when } \dot{\Phi} > 0 \\ k(x) - \beta\dot{\Phi} & \dot{\Phi} < 0 \end{cases}$$

$$\Rightarrow \dot{s} = \begin{cases} \left(f - \dfrac{b}{\hat{b}}\hat{f}\right) + \left(1 - \dfrac{b}{\hat{b}}\right)\left(-\ddot{x}_d + \lambda\dot{\tilde{x}}\right) - \dfrac{b}{\hat{b}}\left(k - \dfrac{\dot{\Phi}}{\beta}\right)\mathrm{sat}\left(\dfrac{s}{\Phi}\right) \\ \quad // \quad - \dfrac{b}{\hat{b}}\left(k - \beta\dot{\Phi}\right)\mathrm{sat}\left(\dfrac{s}{\Phi}\right) \end{cases}$$

Inside the band. $|s| \leq \Phi$ (decay of s)

$$\Rightarrow \dot{s} = \quad // \quad - \frac{b}{\hat{b}}\,\bar{k}\,\frac{s}{\Phi}$$

we define.

$$\frac{\bar{k}(x_d)}{\Phi}\underbrace{\left(\frac{b}{\hat{b}}\right)_{max}}_{=\beta_d} := \lambda.$$

Balance law

$$\bar{k}(x_d) = \frac{\lambda\Phi}{\beta_d}.$$

Dynamics of Φ :

by definition of $\bar{k}(x)$

$$\dot{\Phi} > 0 \qquad \frac{\lambda\Phi}{\beta_d} = k(x_d) - \frac{\dot{\Phi}}{\beta_d}$$

$$\dot{\Phi} < 0. \qquad \frac{\lambda\Phi}{\beta_d} = k(x_d) - \beta_d\dot{\Phi}$$

$$\beta_d \bar{k}(x_d) = \lambda \Phi = \lambda^n \varepsilon \qquad \left(\frac{\Phi}{\lambda^{n-1}} := \varepsilon \right)$$

$(\text{band width})^n \times (\text{tracking precision})$

$\approx (\text{uncertainty along desired trajectory})$

$|s| \leq \Phi \Rightarrow |x - x_d| \leq \frac{\Phi}{\lambda^{n-1}}$

§ 7.3. The Modeling / Performance Trade_Offs

$\lambda^n \varepsilon \approx \beta_d k_d.$

$|\hat{f} - f| \leq F \Rightarrow k = F + \eta$

$\Delta \varepsilon \approx \Delta \left(\frac{\beta_d k_d}{\lambda^n} \right)$

* Better knowledge of $\underbrace{\text{uncertainty}}_{f, b}$ $\Rightarrow$ better precision

* If λ large $\Rightarrow$ do not put much efforts
to reduce β_d, k_d.

Key : How large λ can be chosen,

i) structural resonant $\quad \lambda \leq \frac{2\pi}{3} \nu_R$

ii) neglected time delay $\quad \lambda \leq \frac{1}{3 T_A}$

} Hardware

iii) sampling rate $\quad \lambda \leq \frac{1}{5} \nu_s$ $\quad$ · Software

i) motor $\qquad$ ii) actuator $\qquad$ iii) computation time

§ 7.4 Multi input System

$$x_i^{(n_i)} = f_i(\mathbf{x}) + \sum_{j=1}^{m} b_{ij}(\mathbf{x}) u_j \qquad 1 \le i, j \le 2$$

Assume

$$|\hat{f}_i - f_i| \le F_i$$

$$B = (I + \Delta)\hat{B}, \qquad |\Delta_{ij}| \le D_{ij}, \qquad B, \hat{B} : \text{invertible}$$

$$D_{ii} < 1.$$

Define

$$s_i := \left(\frac{d}{dt} + \lambda_i\right) \tilde{x}_i$$

Want to make

$$\frac{1}{2} \frac{d}{dt} s_i^2 \le -\eta_i |s_i|$$

$$\dot{s}_i = \ddot{\tilde{x}}_i + \lambda_i \dot{\tilde{x}}_i$$

$$= f_i(\mathbf{x}) + \sum_{j=1}^{m} b_{ij}(\mathbf{x}) u_j - \ddot{x}_{d_i} + \lambda_i \dot{\tilde{x}}_i$$

$$u := \hat{B}^{-1}\left(\ddot{x}_d - \hat{f} - \lambda\dot{\tilde{x}} - k\, \mathrm{sgn}(s) \right)$$

$$\Rightarrow \dot{s}_i = f_i(\mathbf{x}) - \ddot{x}_{d_i} + \lambda_i \dot{\tilde{x}}_i + (\delta_{ij} + \Delta_{ij})\left(\ddot{x}_{d_j} - \hat{f}_j - k_j \mathrm{sgn}(s_j) - \lambda \dot{\tilde{x}}_j \right)$$

$$\le -\eta_i \,\mathrm{sgn}(s_i)$$

We can choose

$$(1 - D_{ii}) k_i + \sum_{i \neq j} D_{ij} k_j = F_i + \sum_{j=1}^{n} D_{ij} \left| \ddot{x}_{dj} - \hat{f}_j \right| + \eta_i$$

$$-\lambda \dot{\tilde{x}}_j$$

Last Note.

$\dot{x}_1 = x_2$

$\dot{x}_2 = \cancel{f+} u.$

Matched uncertainty

SMC : good.

$\dot{x}_1 = x_2 + f$

$\dot{x}_2 = x + u.$

Mis matched uncertainty

Integral SMC